

## DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

In Collaboration with



Department of CS & IT



Department of AI & DS

# VISHWAGYAN – INNOVISTA

“Universal Knowledge”

A Real-World Technology Innovation & Solution Showcase



Exam Preparation App with Adaptive Quizzes  
Team 3 | Section 7 | Guide: Dr. Y. Prasanthi  
Course: Database Systems Engineering & Distributed Backend Development



Date:  
03 October 2026



Time:  
09:00 AM – 04:00 PM



Venue:  
Lab 15



### WHAT IS EXAM PREP APP?

An Android app giving every learner personalized exam preparation. Item Response Theory (IRT) adapts question difficulty to each learner's ability. Built using:



Front-End UI (Jetpack Compose)



Backend Service (FastAPI)



REST API Development & Integration



Database Design (Room, Firebase, SQL)



Postman API Testing



ACID Transactions & Indexing



Local + Cloud Data Synchronization



Adaptive Learning with IRT



### EXPECTED OUTCOMES

Learners will be able to:

- ✓ Take adaptive quizzes tuned to their ability using Item Response Theory (IRT)
- ✓ Practice a subject-wise question bank
- ✓ Track performance with analytics
- ✓ Detect weak topics automatically
- ✓ Schedule and attempt mock tests
- ✓ Compare progress via peer ranking
- ✓ Store data reliably using ACID transactions in normalized SQL
- ✓ Sync Room (local) and Firebase (cloud)
- ✓ Get faster queries through database indexing and optimization
- ✓ Demonstrate a complete, working end-to-end Android application
- ✓ Learn efficiently with personalized, user-friendly exam preparation



### KEY DELIVERABLE

Each team will demonstrate:



EXAM PREP PROBLEM



JETPACK COMPOSE UI



FASTAPI BACKEND



ROOM, FIREBASE, SQL



QUIZ & RANKING API



POSTMAN TESTING



INTEGRATED WORKING SOLUTION



### STUDENT PARTICIPATION CALL

#### PROBLEM STATEMENT



Traditional exam-prep apps give every learner the same questions, ignoring knowledge level and performance, causing inefficient learning.

**Don't just learn technology –  
BUILD with it!**

Proposed: an IRT-based adaptive quiz engine with analytics, weak-topic detection, mock tests and peer ranking for personalized prep.

**Come with your idea. Build it. Test it.  
Demonstrate it.**

**LET YOUR SOLUTION SPEAK FOR YOU!**



### PROJECT TEAM MEMBERS

Team No: 3 | Section: 7



Konda Vaishali

Roll No: 2520030117  
Team Member



P. Bindhu Madhavi

Roll No: 2520030077  
Team Member



Alaparathi Nava Harsha

Roll No: 2520030493  
Team Member



Dr. Y. Prasanthi

Faculty  
Project Guide



### INSTRUCTORS / SECTIONS

1. Dr. M. Purushotham
2. Dr. R. Sateesh Kumar
3. Dr. S. Spandana
4. Dr. Y. Subbarayudu
5. Dr. D. Ramya Krishna

6. Dr. Y. Prashanthi
7. Dr. S. K. Khaja Shareef
8. Dr. M. Gayatri
9. Dr. Divya Priya



COURSE COORDINATOR  
Dr. R. Sateesh Kumar



HOD – CSE  
Dr. Venkateshwara Rao



PRINCIPAL  
Dr. L. Koteswara Rao



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From Real-World Problems to Working Digital Solutions!